Intrusion Prevention System (IPS)

LetsDefend SOC Analyst lesson | Asha Latha Panuganti | October 2026

Summary

Hands-on lab analysing Suricata alerts to decide what the traffic was, which attack or tool triggered the signature, and whether the attack succeeded.

Objectives

- Understand how an IPS differs from an IDS: it can act on traffic, not only alert.
- Extract IPs, ports, signatures and categories from Suricata JSON alerts and judge the outcome.

Environment and tools

LetsDefend lab VM (Windows), Notepad++, Suricata EVE-style JSON alert logs.

Method

- Opened each alert log in Notepad++ and read the fields `src_ip`, `dest_port`, `signature`, `category`, `severity` and `action`.
- Used the signature name and category to decide what the alert meant.
- Checked flow, HTTP status and TLS data to judge whether the attack succeeded.

Findings

Log	Key fields observed	Finding
Suricata 1	192.168.59.71:80 to 192.168.59.81:56040; signature "ET ATTACK_RESPONSE Net User Command Response"; category Successful User Privilege Gain; severity 1	Alert is on the **response** to a Windows *net user* command, so the command **ran successfully** and the attacker gained command execution.
Suricata 2	10.60.11.10:443 to 10.60.11.20; TLS SSLv3; signature "ET POLICY SSLv3 inbound connection to server vulnerable to POODLE attack"; action allowed	Attempt against the **POODLE** SSLv3 weakness. The connection was allowed, not blocked.
Suricata 3	192.168.113.45 to 192.168.113.55:80; signature "ET SCAN Possible Nmap User-Agent Observed"; user agent contains Nmap Scripting Engine; GET; status 401; action allowed	Reconnaissance with **Nmap** (NSE scripts). The server answered 401, so the scan did not reach protected content.

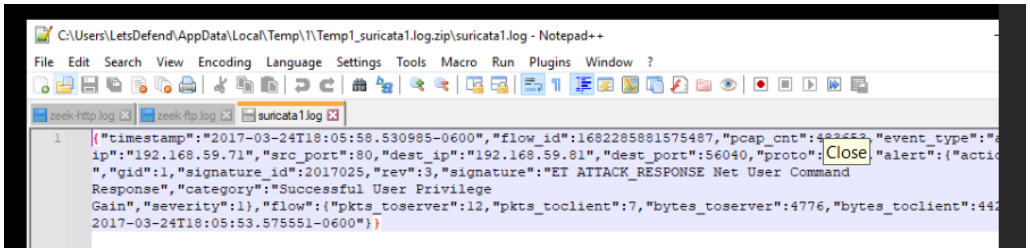


Figure 1: Suricata alert for a successful Net User command response.

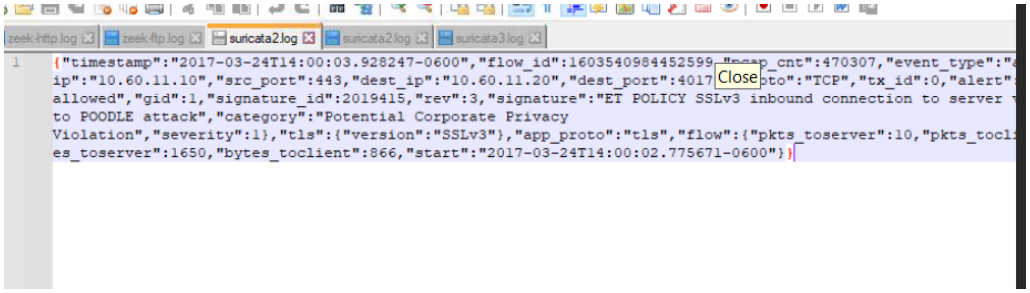


Figure 2: Suricata alert for SSLv3 / POODLE.

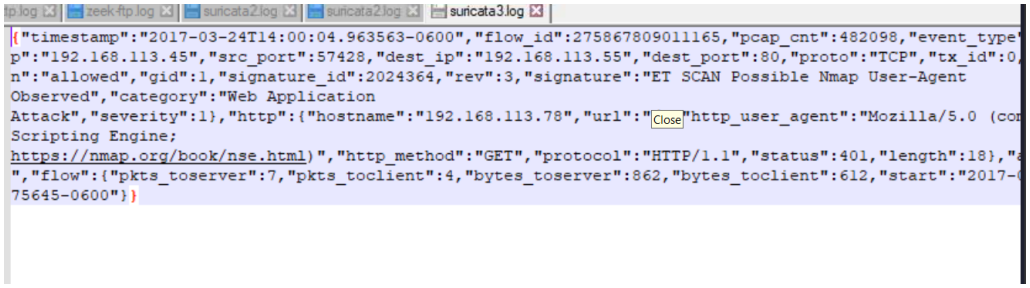


Figure 3: Suricata alert for an Nmap scan.

Security observations

- **Alert does not mean blocked.** The POODLE and Nmap alerts show action "allowed". An IPS only protects when rules are set to drop, so these signatures need drop rules and tuning.
- **Category matters for triage.** "Successful User Privilege Gain" at severity 1 means the attack worked and should be escalated and the host isolated, unlike a scan alert.
- **SSLv3 is obsolete.** The fix is to disable SSLv3 and use TLS 1.2 or later.
- **User-agent detection is weak.** Nmap can spoof its user agent, so scan detection should also use behaviour such as many ports or requests in a short time.

Skills demonstrated

Alert triage, network protocols (HTTP, TLS), IPS/IDS concepts, reading JSON alerts, documenting findings.

Next steps

- Write a custom Suricata rule and set it to drop, then test it against a PCAP.
- Send Suricata alerts to Wazuh and build a dashboard of signatures, categories and actions.